

ARRAY FUNCTIONS IN DATA SCIENCE USING NUMPY

AIM

To understand array creation, reshaping, indexing and slicing using the NumPy library.

PROGRAM

```
import numpy as np

# array = np.random.randint(1, 100, 9) # 9 random integers between 1 and 99
array = np.array([83, 25, 19, 47, 62, 15, 96, 39, 51])
print("Array          :", array)
print("Square root     :", np.sqrt(array))
print("Dimensions (ndim) :", array.ndim)

new_array = array.reshape(3, 3)
print("\nReshaped to 3 x 3:\n", new_array)
print("ndim after reshape :", new_array.ndim)
print("Shape              :", new_array.shape)
print("ravel()            :", new_array.ravel())

newm = new_array.reshape(3, 3)
print("\nnewm[2, 1:3]       :", newm[2, 1:3])
print("newm[1:2, 1:3]     :", newm[1:2, 1:3])
print("new_array[0:3, 0:0]:", new_array[0:3, 0:0])
print("new_array[0:2, 0:1]:\n", new_array[0:2, 0:1])
print("new_array[0:3, 0:1]:\n", new_array[0:3, 0:1])
print("new_array[1:3]      :\n", new_array[1:3])

print("\nSum              :", new_array.sum())
print("Mean              :", new_array.mean())
print("Maximum / index    :", new_array.max(), "/", new_array.argmax())
print("Row-wise sum       :", new_array.sum(axis=1))
print("Column-wise sum    :", new_array.sum(axis=0))
print("Transpose         : \n", new_array.T)
```

OUTPUT

```
Array          : [83 25 19 47 62 15 96 39 51]
Square root     : [9.11043358  5.          4.35889894  6.8556546   7.87400787  3.87298335
 9.79795897
 6.244998   7.14142843]
Dimensions (ndim) : 1

Reshaped to 3 x 3:
[[83 25 19]
 [47 62 15]
 [96 39 51]]
ndim after reshape : 2
Shape              : (3, 3)
ravel()            : [83 25 19 47 62 15 96 39 51]

newm[2, 1:3]       : [39 51]
newm[1:2, 1:3]     : [[62 15]]
new_array[0:3, 0:0]: []
new_array[0:2, 0:1]:
[[83]
 [47]]
new_array[0:3, 0:1]:
[[83]
 [47]
 [96]]
new_array[1:3]      :
```

```
[[47 62 15]
 [96 39 51]]
```

```
Sum          : 437
Mean         : 48.55555555555556
Maximum / index : 96 / 6
Row-wise sum  : [127 124 186]
Column-wise sum : [226 126 85]
Transpose    :
[[83 47 96]
 [25 62 39]
 [19 15 51]]
```

RESULT

Thus the NumPy array functions (creation, reshape, ravel, slicing and aggregation) were executed and the output was verified.